

Higher SNR or Lower Latency? Investigating the Effects of SNR and Latency on Hearing in Noise with Single-sided Deafness

Ken Takaki
The University of Tokyo
Bunkyo-ku, Tokyo, Japan
takaki@akg.t.u-tokyo.ac.jp

Yumi Okano
Department of Speech-Language-Hearing Therapy
Gunma Paz University
Takasaki-shi, Gunma, Japan
okano@paz.ac.jp

Kouei Yamaoka
The University of Tokyo
Bunkyo, Tokyo, Japan
kouei_yamaoka@ipc.i.u-tokyo.ac.jp

Yoshihiro Kawahara
The University of Tokyo
Bunkyo-ku, Tokyo, Japan
kawahara@akg.t.u-tokyo.ac.jp

Abstract

Hard-of-hearing restricts social participation in hearing/speaking cultures due to the difficulty hearing in noisy situations. Listening in severely noisy environments remains challenging even with hearing aids. Deep learning-based high-performance noise reduction methods have advanced, but these typically introduce significantly higher signal processing delays compared to traditional hearing aids. It is unclear whether such high-latency, high-performance signal processing would improve audibility for hard-of-hearing individuals. Therefore, this study conducted experiments to investigate the impact of audio delay and signal-to-noise ratio (SNR) on sound perception and conversation. Among the broad spectrum of hard-of-hearing, we focused on single-sided deafness based on the low benefit from conventional hearing aids due to insufficient SNR. Our results show that word recognition accuracy improved by 24% with 6 dB SNR improvement, allowing 400 ms additional delay. Furthermore, in conversation experiments, responses were sometimes faster even when the user heard the partner's voice with a 400 ms delay, if the SNR was 6 dB better. These findings suggest that sacrificing latency, traditionally considered crucial in hearing aids, for improved SNR in signal processing can lead to a better conversation experience for hard-of-hearing individuals.

CCS Concepts

• **Human-centered computing** → **Accessibility technologies**; *Sound-based input / output; Accessibility systems and tools.*

Keywords

single-sided deafness, hearing aids, assistive technology, auditory perception

ACM Reference Format:

Ken Takaki, Kouei Yamaoka, Yumi Okano, and Yoshihiro Kawahara. 2025. Higher SNR or Lower Latency? Investigating the Effects of SNR and Latency

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

UIST Adjunct '25, Busan, Republic of Korea

© 2025 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-2036-9/25/09

<https://doi.org/10.1145/3746058.3758368>

on Hearing in Noise with Single-sided Deafness. In *The 38th Annual ACM Symposium on User Interface Software and Technology (UIST Adjunct '25)*, September 28–October 01, 2025, Busan, Republic of Korea. ACM, New York, NY, USA, 3 pages. <https://doi.org/10.1145/3746058.3758368>

1 Introduction and Related Work

Hard-of-hearing individuals have difficulty hearing in noisy environments [11], which severely impedes their social participation in speaking cultures [9]. Hearing aids are often perceived as ineffective in very noisy social situations where the signal-to-noise ratio (SNR) is low, considering conversational speech as signal and irrelevant speakers or environmental sounds as noise [14].

To enhance the performance of hearing aids, leveraging low-latency (<20 ms) deep learning-based signal processing methods [15, 16] is a promising approach. However, the low-latency signal processing model used in these works is estimated to have an 8.1 dB lower SNR performance compared to offline methods, which do not have real-time latency constraints [2, 18]. Additionally, diffusion-based noise reduction techniques have been reported to achieve performance comparable to offline methods by tolerating algorithmic delays of 320 ms or more [8]. These results suggest that relaxing the latency requirement to several hundred milliseconds can result in a significant reduction of noise.

Given that current low-latency audio processing offers insufficient noise reduction for hard-of-hearing individuals, we propose using higher-performance noise reduction by tolerating increased latency in hearing aids.

This approach requires evaluating its impact from three perspectives: (a) whether delayed audio affects speech production, (b) whether it affects listening comprehension, and (c) whether it affects conversational turn-taking. Regarding (a), hearing one's voice with a delay of more than 30 ms starts to inhibit speech, suggesting that hearing aid delays should be under 20 ms [12]. However, using own-voice activity detection to mute the hearing aid output when the user is speaking could potentially relax this constraint [6]. For (b), speech intelligibility starts to decline when audio is delayed by 180 ms to over 400 ms relative to visuals [3]. Concerning (c), experiments on video calls show that a 100 ms delay is acceptable for turn-taking, but a 700 ms delay causes significant disruption [10]. However, existing experiments related to (b) and (c) were conducted

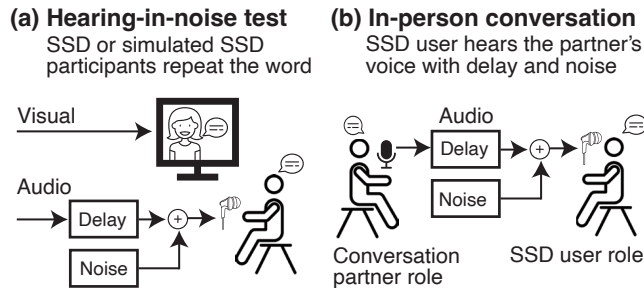


Figure 1: Experiments conducted in our study. (a) Hearing-in-noise tests present words visually with delayed audio plus noise. (b) In-person conversation experiment in which the SSD user hears the sound with a delay and noise to simulate using high-delay hearing aids in noisy environments.

in environments with little or fixed noise levels. Therefore, the optimal balance between acceptable delay and improved SNR remains unknown.

To address this gap, as shown in Figure 1, this study evaluated the impact of varying delay and SNR from two perspectives: a word intelligibility test to assess listening ability and a two-person conversation experiment to qualitatively understand the effects on conversation. Among individuals with hard-of-hearing, we focus on those with single-sided deafness (SSD), who have one hearing ear, because they benefit less from conventional hearing aids due to insufficient SNR [1]. Experiments demonstrated that higher SNR with audio delay yielded higher word comprehension scores than lower SNR without audio delay. Our results suggest that high-performance audio processing, which may introduce greater latency but offers superior noise reduction, could be more valuable for SSD users and other hard-of-hearing individuals, compared to current hearing aid signal processing optimized primarily for minimal delay.

2 Hearing in Noise Experiment

We investigated the effects of audio delay and SNR on word intelligibility by measuring word recognition accuracy.

Experiment Setup. As shown in Figure 1 (a), sound stimuli were presented via Etymotic Research ER-2 earphones. For SSD participants, the earphone was in their better-hearing ear. For hearing participants, a simulated SSD condition was created by using an earphone in the left ear and an earplug in the right.

Stimuli. Based on preliminary tests, we used three challenging SNR conditions: -12 dB, -6 dB, and 0 dB. Audio delay conditions relative to visual stimuli were 0 ms, 400 ms, and “no video,” totaling nine unique combinations (3 SNR × (2 delay + no video)). Audio and video materials came from clinically validated speech intelligibility tests [5], comprising three of five 50-word phonetically balanced lists. Presentation order of the lists was randomized. Speech-shaped noise [4] was used as background noise, adjusted to 65 dB SPL, a common level in hearing-in-noise tests [13]. Participants repeated words they heard; only perfect matches were scored as correct.

Results. Two authors with SSD and three normal-hearing individuals took part with informed consent. Figure 2 displays average word recognition accuracy and standard deviation. A Wilcoxon

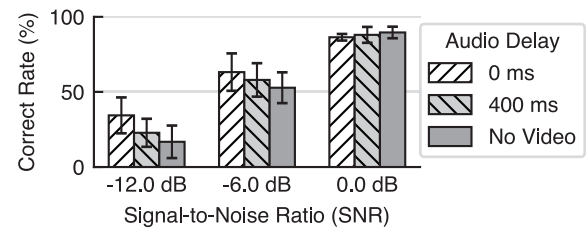


Figure 2: Average word correct rate and standard deviation across different SNR and delay conditions.

signed-rank test ($p < 0.05$) showed significantly higher accuracy for SNR = -6 dB, delay = 400 ms ($M=58.0\%$, $SD=11.1\%$) than SNR = -12 dB, delay = 0 ms ($M=34.4\%$, $SD=11.9\%$, $p=0.031$). Likewise for SNR = 0 dB, delay = 400 ms ($M=88.0\%$, $SD=5.3\%$) and SNR = -6 dB, delay = 0 ms ($M=63.2\%$, $SD=12.5\%$, $p=0.031$). This indicates a 6 dB SNR improvement significantly boosts accuracy even with a 400 ms audio delay. Regarding the impact of audio delay and the presence/absence of video on accuracy at the same SNR, audio delay tended to decrease accuracy at lower SNRs. Moreover, the condition without video showed the lowest accuracy at low SNRs.

3 In-person Conversation Experiment

Next, we investigated the impact of speech with increased SNR and additional latency on actual conversation, exploring both user and conversational partner perspectives.

Experiment Setup and Stimuli. Two participants took part: one assumed the role of an SSD user who utilizes a noise reduction system due to difficulty hearing in noise, and the other played the role of a conversational partner. As shown in Figure 1 (b), the probe earphone presents the other participant’s voice captured by a SHURE SM35-XLR microphone and 65 dB SPL speech-shaped noise to the SSD user to simulate the situation. The conversation partner can hear the SSD user’s voice directly. Because strict SNR control in natural conversation is difficult, for the “No Latency” (N) condition, the microphone gain was adjusted to require significant listening effort. The “Latency Aid” (L) condition added a 400 ms delay to the voice and increased the microphone gain by 6 dB to increase SNR. The conversation lasted for three minutes, talking about their activities during their student days.

Results. Two authors with SSD underwent the experiment. A1, a speech therapist, was interviewed by A2. A1 was asked about listening ease and conversational flow from the “user” perspective. A1 was also asked if the SSD user was unnatural, specifically regarding response and nodding timing from the “conversational partner’s” perspective. From the user perspective A1 stated: “In the (L) condition, the other person’s voice sounded louder. However, I found the (N) condition easier to hear.” A1 also noted, “If I couldn’t see their mouth, it might be easier to hear in the (L) condition” but expressed anxiety about not seeing the mouth. From the conversational partner’s perspective, A1 stated: “(the user’s response) did not seem unnatural or that miscommunication was occurring in the (L) condition. However, in the (N) condition, (the user) seemed to be trying harder to listen.” A1 further stated that “When I asked a sudden question in the (L) condition, I felt the response came back quickly.”

This experiment revealed individual differences: one preferred synchronized mouth movements (N), while the other seemed to hear easily with higher SNR and 400 ms latency (L). On the other hand, the conversational partner did not perceive delayed responses of the SSD user with the Latency Aid. Moreover, responses appeared faster with Latency Aid, suggesting improved intelligibility led to quicker responses. This observation is supported by literature showing that both normal-hearing and SSD individuals respond faster to speech with less noise [7, 17].

4 Conclusion and Future Work

This preliminary study investigated the effect of audio delay and SNR on listening and conversation. Hearing-in-noise tests showed that higher SNR, even with increased delay, yielded superior word comprehension. Our findings suggest that sacrificing latency, considered crucial in hearing aids, for improved SNR in signal processing can enhance the conversation experience for hard-of-hearing individuals. In future work, we will further evaluate conversations objectively/subjectively under diverse SNR and delay conditions, including participants with bilateral hearing loss.

Acknowledgments

This work was supported by JSPS KAKENHI Grant Number JP25K03149.

References

- [1] Coosje Jacoba Isabella Caspers, Rik Chrétien Nelissen, Hans J M M Groenewoud, and Myrthe Karianne Sophie Hol. 2022. Hearing-related quality of life in 75 patients with a percutaneous bone conduction device. *Otology & Neurotology* 43, 3 (March 2022), 345–351.
- [2] Guo Chen, Kai Li, Runxuan Yang, and Xiaolin Hu. 2025. Time-frequency-based attention cache memory model for real-time speech separation. *arXiv [cs.SD]* (May 2025).
- [3] Milton Chen. 2003. A low-latency lip-synchronized videoconferencing system. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (CHI '03)*. Association for Computing Machinery, New York, NY, USA, 465–471.
- [4] Wouter A Dreschler, Hans Verschuure, Carl Ludvigsen, and Soren Westermann. 2001. ICRA noises: artificial noise signals with speech-like spectral and temporal properties for hearing instrument assessment. *International Collegium for Rehabilitative Audiology. Audiology* 40, 3 (May 2001), 148–157.
- [5] Yumiko Fukuda and Akira Yokkaichi. 1992. Evaluating auditory and visual speech perception ability in the hearing impaired. *The Japan Journal of Logopedics and Phoniatrics* 33, 2 (April 1992), 177–185.
- [6] Poul Hoang, Zheng-Hua Tan, Thomas Lunner, Jan Mark de Haan, and Jesper Jensen. 2020. Maximum likelihood estimation of the interference-plus-noise cross power spectral density matrix for own voice retrieval. In *ICASSP 2020 - 2020 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 6939–6943.
- [7] Zahra Jafari, Hardeep Ryait, Mohammad Habibnezhad, Claire Niehaus, Erin Cantwell, Jordan Dudley, Bryan Kolb, and Majid Mohajerani. 2025. Reaction time, speech recognition, and verbal memory performance: Nonnative versus native English speakers. *Journal of Speech, Language, and Hearing Research: JSLHR* (June 2025), 1–13.
- [8] Bunlong Lay, Rostilav Makarov, and Timo Gerkmann. 2025. Diffusion buffer: Online diffusion-based speech enhancement with sub-second latency. *arXiv [eess.AS]* (June 2025).
- [9] Laura Lucas, Roulla Katiri, and Pádraig Thomas Kitterick. 2018. The psychological and social consequences of single-sided deafness in adulthood. *International Journal of Audiology* 57, 1 (Jan. 2018), 21–30.
- [10] Lucas M Seuren, Joseph Wherton, Trisha Greenhalgh, and Sara E Shaw. 2021. Whose turn is it anyway? Latency and the organization of turn-taking in video-mediated interaction. *Journal of Pragmatics* 172 (Jan. 2021), 63–78.
- [11] Guido F Smoorenburg. 1992. Speech reception in quiet and in noisy conditions by individuals with noise-induced hearing loss in relation to their tone audiogram. *The Journal of the Acoustical Society of America* 91, 1 (Jan. 1992), 421–437.
- [12] Michael A Stone and Brian C J Moore. 2002. Tolerable hearing aid delays. II. Estimation of limits imposed during speech production. *Ear and Hearing* 23, 4 (Aug. 2002), 325–338.
- [13] Ken Takaki, Akinori Kashio, Etsushi Nozaki, Tomomi Kanai, Teru Kamogashira, Fumikazu Saze, Tsukasa Uranaka, Shinji Urata, Hajime Koyama, Yoshihiro Kawahara, and Tatsuya Yamasoba. 2024. A randomized crossover study in single-sided deafness comparing a cartilage conduction CROS system and an air-conduction CROS system. *Otology & Neurotology* 45, 6 (July 2024), 635–642.
- [14] Ken Takaki, Etsushi Nozaki, Tomomi Kanai, Ari Hautasaari, Akinori Kashio, Daisuke Sato, Teru Kamogashira, Tsukasa Uranaka, Shinji Urata, Hajime Koyama, Tatsuya Yamasoba, and Yoshihiro Kawahara. 2023. asEars: Designing and Evaluating the User Experience of Wearable Assistive Devices for Single-Sided Deafness. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems (CHI '23, Article 44)*. Association for Computing Machinery, New York, NY, USA, 1–17.
- [15] Bandhav Veluri, Malek Itani, Justin Chan, Takuya Yoshioka, and Shyamnath Gollakota. 2023. Semantic Hearing: Programming Acoustic Scenes with Binaural Hearables. In *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology (UIST '23, Article 89)*. Association for Computing Machinery, New York, NY, USA, 1–15.
- [16] Bandhav Veluri, Malek Itani, Tuochao Chen, Takuya Yoshioka, and Shyamnath Gollakota. 2024. Look Once to Hear: Target Speech Hearing with Noisy Examples. In *Proceedings of the CHI Conference on Human Factors in Computing Systems (CHI '24, Article 37)*. Association for Computing Machinery, New York, NY, USA, 1–16.
- [17] Marcus Windsor Rao Voola, Dayse Tavora-Vieira, Andre Wedekind, Caris Boganov, and Aanand Acharya. 2024. Auditory listening effort and reaction time: a comparative study between single sided deaf cochlear implant users and normal hearing controls. *Frontiers in Audiology and Otology* 2 (July 2024), 1369812.
- [18] Zhong-Qiu Wang, Samuele Cornell, Shukjae Choi, Younglo Lee, Byeong-Yeol Kim, and Shinji Watanabe. 2023. TF-GridNet: Integrating full- and sub-band modeling for speech separation. *ACM Transactions on Audio, Speech, and Language Processing* 31 (Aug. 2023), 3221–3236.